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MemSFT: Mitigating Alignment Tax with an External Parametric Memory

A Plug-and-Play Memory Module Imparts Domain Knowledge Without Touching Backbone Weights—Matching Full SFT While Preserving General Capability

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Fine-tuning an LLM on a specialized domain reliably degrades its general-purpose capability—a phenomenon called the alignment tax. **MemSFT** avoids this by never updating backbone parameters: a compact parametric memory module is trained to imitate a domain retriever, and a learned router fuses memory and backbone distributions token-by-token at inference. Across biology, geoscience, and law, MemSFT exceeds full SFT on domain tasks while degrading general benchmarks by only 0.2%—versus a 20.5-point drop for full SFT.

AT A GLANCE

Problem: Fine-tuning on domain data causes catastrophic forgetting of general capabilities, limiting the practical utility of LLM adaptation.

Why it matters: Deploying LLMs in law, medicine, and science requires domain expertise without sacrificing general reasoning.

Method: Train a small parametric memory to distill a domain retriever; a router dynamically fuses memory and backbone at each decoding step.

Result: 72.6 domain avg. (vs. 71.8 for full SFT); 74.6 general avg. (vs. 54.3 for full SFT).

Evidence: Evaluated on biology, geoscience, law with Qwen3-8B through Qwen3-235B-A22B; memory transfers across model sizes.

The Big Picture

Large language models have transformed the landscape of domain-specific AI: a single pretrained model can serve as the foundation for law, medicine, finance, and science—but only with significant customization. Full supervised fine-tuning (SFT) on domain data is the dominant adaptation method, but it carries a well-documented cost: catas-

trophic forgetting. General benchmarks drop 15–30% relative to the base model after full SFT on a specialized corpus.

MemSFT recasts domain adaptation as a knowledge injection problem. Instead of modifying the backbone, it adds a compact module that holds the domain knowledge. The backbone remains intact; the memory supplies the domain delta; a router blends them conditionally during generation.

Why Existing Approaches Fall Short

Three adaptation paradigms exist, each with a critical limitation:

Full SFT: Updates all backbone parameters. Achieves strong domain performance but causes catastrophic forgetting (general benchmarks drop 15–30%). Once fine-tuned, the backbone cannot easily serve multiple domains simultaneously.

LoRA/adapters: Updates low-rank adapter matrices ($\ll 1\%$ of parameters). Reduces forgetting but does not eliminate it; domain gains are weaker than full SFT.

Retrieval-augmented generation (RAG): Retrieves relevant documents at inference time. Preserves general capability but adds retrieval latency ($\sim 100\text{--}500$ ms per query), requires a maintained document store, and does not internalize patterns that allow efficient pattern completion.

MemSFT achieves full SFT domain performance with RAG-level general capability preservation, and with only 15% additional inference latency over a vanilla backbone—no retrieval system required.

Core Mechanism

MemSFT consists of two components trained sequentially without backbone modification:

Parametric memory M_ϕ : A compact transformer ($\sim 75\text{M}$ parameters) trained to imitate a dense retriever operating over domain documents. Given the generation prefix $h_{1:t}$, M_ϕ produces a distribution over the next token that reflects domain-relevant patterns.

Router g_ψ : A lightweight gating network ($\sim 1\text{M}$ parameters) that predicts, at each decoding step, what fraction of the output distribution should come from the memory vs. the backbone.

Technical Formulation

The memory is trained by knowledge distillation from a retriever:

$$\mathcal{L}_{\text{mem}} = D_{\text{KL}}(\text{Retriever}(\cdot \mid h_{1:t}, \mathcal{D}_{\text{dom}}) \parallel M_\phi(\cdot \mid h_{1:t}))$$

where $\mathcal{D}_{\text{dom}}$ is the domain corpus. The retriever is a dense retrieval model that augments a frozen LM with top- k documents.

The final token distribution is:

$$p(w_t | w_{<t}) = g_\psi(w_{<t}) \cdot M_\phi(w_t | w_{<t}) + (1 - g_\psi(w_{<t})) \cdot p_\theta(w_t | w_{<t})$$

where p_θ is the frozen backbone and $g_\psi(w_{<t}) \in [0, 1]$.

The router is trained to maximize the likelihood of domain data under $p(\cdot)$, with a sparsity regularizer:

$$\mathcal{L}_{\text{router}} = -\mathbb{E}_{(w,c) \sim \mathcal{D}_{\text{dom}}} [\log p(w | c)] + \lambda \|g_\psi\|_1$$

Learning or Inference Procedure

Training proceeds in two stages:

Stage 1 (memory distillation): Train M_ϕ against the retriever output distribution on domain documents for 10K steps with Adam. Backbone is frozen throughout.

Stage 2 (router training): Freeze M_ϕ , train g_ψ on a held-out domain set for 1K steps. The sparsity regularizer encourages the router to invoke memory selectively: on non-domain prompts, the router converges to $g_\psi \approx 0$ (pure backbone).

At inference, the backbone and memory run in parallel; the router blends their outputs. Per-step latency overhead is $\approx 15\%$ for Qwen3-8B.

What the Guarantee Says

Since the backbone parameters θ are never modified, the backbone’s general-task performance is preserved exactly by construction. Any degradation in general benchmarks must arise from the router incorrectly invoking domain memory for non-domain prompts. The sparsity regularizer controls this: the expected general degradation is bounded above by $\lambda \cdot \mathbb{E}[g_\psi(w_{<t})]$ on non-domain contexts, which converges to near-zero with appropriate λ .

Experimental Findings

Domains: Biology (PubMedQA, BioASQ), Geoscience (GeoQA), Law (LegalBench). **Models:** Qwen3-8B to Qwen3-235B-A22B.

Method	Domain avg.	General avg.
Base LLM	62.3	74.8
Full SFT	71.8	54.3
LoRA	68.4	70.1
RAG	70.2	74.7
MemSFT	72.6	74.6

MemSFT exceeds full SFT on domain tasks by 0.8 points while preserving general capability (only 0.2-point drop versus a 20.5-point drop for full SFT). It also exceeds RAG by 2.4 domain points with $3\times$ lower inference latency.

Cross-model reuse: A memory trained with Qwen3-8B transfers to Qwen3-72B, recovering 92% of per-domain performance vs. a native Qwen3-72B memory—demonstrating that the memory is architecturally portable across backbone sizes.

Ablations and Interpretation

Router sparsity: Removing $\lambda \|g_\psi\|_1$ causes the router to always blend both distributions ($g_\psi \approx 0.5$), degrading general performance by an additional 3.1 points while providing only marginal domain gains.

Memory size: Scaling from 25M to 100M parameters improves domain performance by 2.4 points; diminishing returns beyond 75M.

Distillation teacher quality: BM25 retriever (sparse) as teacher vs. dense retrieval reduces domain performance by 3.8 points, confirming that retrieval quality is the primary bottleneck for memory fidelity.

Reference

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